UNIT V

Replacement, Maintenance Analysis & DP

Handwritten study notes | B.Tech | Operations Research

Syllabus topics covered in this unit:

- Maintenance types; make-or-buy decision
- Types of replacement problems; economic life of an asset
- Simple probabilistic model: individual and group replacement
- Dynamic programming: Bellman's principle of optimality
- Shortest path, capital budgeting, solving an LP by DP

1. Maintenance - Introduction and Types

Maintenance is the set of activities needed to keep plant, machinery and equipment in good working condition so that production is not interrupted. The OR question is **how much maintenance, and when**, so that the sum of maintenance cost and breakdown loss is minimum.

Types of maintenance

Type	Meaning	Example
Breakdown (corrective)	repair only after failure occurs	bulb, fan, non-critical pump
Preventive	scheduled inspection, lubrication, part change before failure	oil change every 5000 km
Predictive (condition based)	monitor condition (vibration, temperature) and act when needed	bearing vibration analysis
Routine	daily cleaning, lubrication and checks	cleaning of machines
Planned / Scheduled	overhaul at fixed intervals based on records	annual shutdown maintenance

- Preventive maintenance (PM) reduces breakdown cost but increases maintenance cost. Optimum PM level = where **total cost (PM cost + breakdown cost)** is minimum
- PM is worthwhile when the failure rate **increases** with age and breakdown cost is much higher than PM cost

Preventive maintenance - cost trade-off example

A machine shop: each breakdown costs Rs 1500; each preventive maintenance (PM) visit costs Rs 300. Expected breakdowns per month depend on the number of PM visits per month:

PM visits k	Breakdowns	Breakdown cost	PM cost	Total cost
0	10	15000	0	15000
1	6	9000	300	9300
2	4	6000	600	6600
3	3	4500	900	5400
4	2.5	3750	1200	4950
5	2.4	3600	1500	5100

Minimum total cost Rs 4950 at 4 PM visits per month. More PM ($k = 5$) costs more than the breakdowns it saves.

Make or Buy decision

Should a firm **make** a component in-house or **buy** it from a supplier? Compare total cost of both.

Buy: $TC_b = p \times Q$ Make: $TC_m = F + v \times Q$ (F = fixed cost, v = variable cost per unit, p = purchase price)

Break-even quantity $Q_0 = F / (p - v)$: make if $Q > Q_0$, buy if $Q < Q_0$.

Example: buy price Rs 30; in-house fixed cost Rs 1,20,000, variable cost Rs 18/unit. Q_0 = 120000/(30 - 18) = 10,000 units. For requirement 15,000 units: buy = 4,50,000, make = 120000 + 270000 = 3,90,000 -> make. For 6,000 units: buy = 1,80,000, make = 2,28,000 -> buy.

Other factors: quality, capacity, delivery reliability, secrecy, supplier risk, labour and idle capacity.

2. Replacement Problems

Equipment gets worse with use: running (maintenance, repair, operating) cost goes up and resale value goes down. Replacement analysis finds the **best time to replace** so that the cost per unit time is minimum.

Types of replacement problems

- (i) Gradual deterioration: machines, vehicles - running cost increases with age (economic life problem)
- (ii) Sudden failure: electric bulbs, tubes, fuses, transistors - work perfectly till they fail suddenly: **individual vs group replacement** (probabilistic)
- (iii) Staff replacement: people leave/retire; recruitment and training replace them (probability of leaving by year)
- (iv) Obsolescence: equipment replaced by a better design even though working (new technology)

Common terms

- Capital (purchase) cost C; scrap/resale value S; running cost R(t) in year t; cumulative cost; **average annual cost** A(n)
- Economic life: the age n at which the average annual total cost is minimum

3. Economic Life of an Asset (money value ignored)

Average annual cost for n years: $A(n) = [\, C - S_n + \sum_{t=1}^{n} R(t) \,] / n$

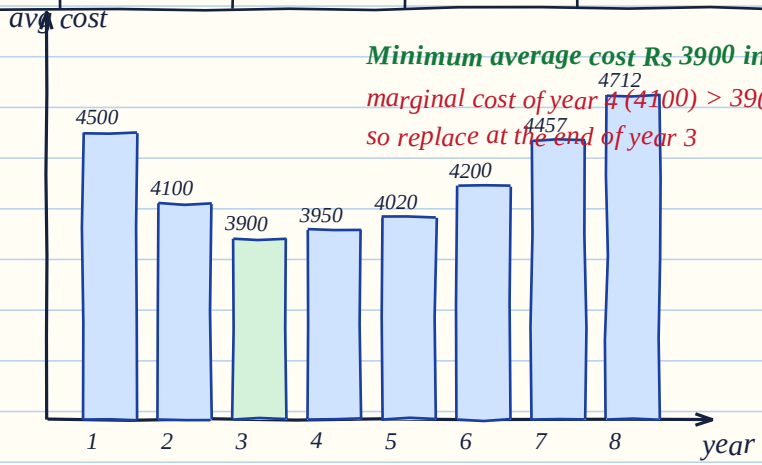
Rule 1: replace at the end of the year n for which A(n) is minimum.

Rule 2 (equivalent): keep the machine as long as the **marginal (next year) cost** $M(n+1) = R(n+1) + S_n - S_{n+1}$ is less than the present average cost A(n); replace when it exceeds A(n).

Example 1

A machine costs Rs 12,000. Running costs and resale values (end of year) are given. When should it be replaced?

Year n	Running cost	Cum. running	Resale S_n	Total cost $P - S_n$ + cum	Average cost	Marginal cost
1	1500	1500	9000	4500	4500	4500
2	1700	3200	7000	8200	4100	3700
3	2000	5200	5500	11700	3900	3500
4	2600	7800	4000	15800	3950	4100
5	3300	11100	3000	20100	4020	4300
6	4100	15200	2000	25200	4200	5100
7	5000	20200	1000	31200	4457	6000
8	6000	26200	500	37700	4712	6500



Minimum average cost Rs 3900 in year 3
marginal cost of year 4 (4100) > 3900
so replace at the end of year 3

Economic life = 3 years. Minimum average cost = Rs 3,900 per year (year 3).
Check by Rule 2: marginal cost in year 4 = 2600 + (5500 - 4000) = Rs 4100 > A(3) = 3900, while year 3's marginal cost 3500 < 3900.

Special case (no resale value, S = 0): replace when the running cost of the next year exceeds the

average total cost up to now. If $R(t) = a + bt$, optimal age $n^* = \sqrt{(2C/b)}$.

Example 1(a): no resale value (scrap = 0)

Machine cost Rs 5000; running costs year 1-7: 500, 700, 900, 1200, 1600, 2100, 2700.

Year	Running	Cumulative	Total (5000 + cum)	Average
1	500	500	5500	5500.0
2	700	1200	6200	3100.0
3	900	2100	7100	2366.7
4	1200	3300	8300	2075.0
5	1600	4900	9900	1980.0
6	2100	7000	12000	2000.0
7	2700	9700	14700	2100.0

Minimum average = Rs 1980 in year 5. Rule check: year 6 running cost 2100 > 1980 (average) so replace at the end of year 5; year 5 running cost 1600 < 1980 so keep up to year 5.

Replacement with money value (discounting)

- Let interest rate be i; discount factor $v = 1/(1 + i)$. Present worth of all costs over the n-year horizon is compared

Annual equivalent cost of keeping for n years: $W(n) = [\ C + \sum_{t=1}^{n} R(t)\, v^{t-1} - S_n\, v^n\ ] / [\ \sum_{t=1}^{n} v^{t-1}\ ]$

Replace at the n which minimises W(n). Rule: replace when the next year's running cost exceeds the weighted average of past costs (interest on capital is counted).

Reason: money available now is worth more than money later, so late-year costs are discounted; the economic life with discounting is usually **longer** than without.

Example 1(b): discounting at 10%

Machine cost Rs 5000; running cost years 1-5: 800, 1000, 1300, 1700, 2200; resale value at end of years 1-5: 3500, 2500, 1800, 1200, 800. Interest 10%, $v = 1/1.1 = 0.9091$.

n	Present worth of cost	$\sum v^{t-1}$	Annual equivalent W(n)
1	2618.2	1.000	2618.2
2	4643.0	1.909	2432.0
3	6431.1	2.736	2350.9
4	8241.1	3.487	2363.5
5	10066.6	4.170	2414.1

Minimum W(n) = Rs 2350.9 at n = 3, so replace every 3 years. (Here running costs are discounted to year-start values and S_n is discounted n years.)

Replacement because of a new, better machine (obsolescence)

Compare **annual cost of the old (defender)** with the minimum average cost of the **new (challenger)**. Replace the old machine now if its next-year cost exceeds the challenger's minimum average cost.

4. Items that Fail Completely - Probabilistic Replacement

Bulbs, tubes, fuses and many electronic parts work till they fail without warning. A **mortality table** gives p_i = probability that an item fails in the i-th period ($\sum p_i = 1$).

- Mean life = $\sum i \cdot p_i$. Average number of failures per period (steady state) = N / mean life
- Individual replacement: replace each item when it fails (cost c_i each)
- Group replacement: replace ALL items at fixed intervals (cost c_g each, $c_g < c_i$) and also replace individual failures in between

Number of failures in each period

$N_0 = N$ (all new at time 0); $N_t = N\,p_t + N_1\,p_{t-1} + N_2\,p_{t-2} + \ldots + N_{t-1}\,p_1$ (renewal equation)
Group policy cost for interval t: $C(t) = [\,N\,c_g + c_i \sum_{k=1}^{t-1} N_k\,] / t$. Choose t with minimum C(t); compare with individual-policy cost per period = $c_i\,N$ / mean life.

Example 2

1000 bulbs. Failure probability per week: 0.10, 0.15, 0.20, 0.25, 0.30 for weeks 1-5. Replacing one bulb on failure costs Rs 5; replacing all together costs Rs 2 per bulb. Find the best policy.

Week i	1	2	3	4	5
p_i	0.10	0.15	0.20	0.25	0.30

Mean life = 1(0.1) + 2(0.15) + 3(0.2) + 4(0.25) + 5(0.3) = 3.5 weeks. **Individual policy:** failures per week = 1000/3.5 = 285.7; cost = 285.7 × 5 = Rs 1428.6 per week.

Failures in successive weeks: $N_1 = 1000(0.10) = 100$; $N_2 = 1000(0.15) + 100(0.10) = 160$; $N_3 = 1000(0.20) + 100(0.15) + 160(0.10) = 231$; $N_4 = 317.1$; $N_5 = 423.4$.

Group interval t (weeks)	Cum. failures before t	Total cost Rs	Cost per week Rs
1	0	2000	2000.0
2	100	2500	1250.0
3	260	3300	1100.0
4	491	4455	1113.8
5	808	6040	1208.1

Best group policy: replace ALL bulbs every 3 weeks (cost Rs 1100 per week, the minimum).

Individual policy costs Rs 1428.6 per week, so group replacement saves 1428.6 - 1100 = Rs 328.6 per week. Interval 4 gives 1113.75 (higher), so t = 3.

Rule: the optimum group interval is the **last t for which the cost per period decreases**, i.e. replace when the cost of the individual policy for the next period ($c_i N_t$...) exceeds the average cost to date.

Example 3: another mortality table

N = 2000 items; p = 0.05, 0.15, 0.30, 0.50 (weeks 1-4); individual cost Rs 4, group cost Rs 1.5 per item. Mean life = 0.05 + 0.30 + 0.90 + 2.00 = 3.25 weeks; individual policy = 2000 × 4/3.25 = Rs 2461.5 per week.

$N_1 = 100$; $N_2 = 2000(0.15) + 100(0.05) = 305$; $N_3 = 2000(0.30) + 100(0.15) + 305(0.05) = 630.25$.

Interval t	Failures before t	Cost = 3000 + 4 × failures	Per week
1	0	3000	3000
2	100	3400	1700
3	405	4620	1540
4	1035.25	7141	1785.25

Best: group replacement every 3 weeks (Rs 1540/week) which is much lower than Rs 2461.5/week for individual replacement.

Policy	When to use	Condition
Individual	item fails rarely, group replacement cost is not much less	$c_g \approx c_i$
Group	many identical items, failure rate rises with age	$c_g << c_i$

Staff replacement (idea)

Treat the number of employees leaving in year i as the failure distribution. Using N_t as above, find how many people to recruit and train each year, and compare **group (batch) recruitment** with individual replacement.

Example 4: staff replacement

A firm has 1000 employees. The probability of an employee leaving in year i of service: 0.10, 0.20, 0.30, 0.40 (years 1-4; all leave by year 4). Mean service = 0.1 + 0.4 + 0.9 + 1.6 = 3 years, so in the long run about 1000/3 = 333 recruits per year are needed.

Year t	1	2	3	4
Leavers N_t	100.0	210.0	341.0	506.1

$N_1 = 100$, $N_2 = 1000(0.2) + 100(0.1) = 210$, $N_3 = 300 + 20 + 21 = 341$, $N_4 = 400 + 30 + 42 +$

34.1 = 506.1. The number of leavers oscillates and settles at about 333, so a steady hiring plan of ~333 per year (with batch training) is adopted.

Replacement because of obsolescence - quick example

Old machine: next-year operating cost Rs 4000. New machine: minimum average annual cost Rs 3500 (from its own table). Since 4000 > 3500 the old machine should be replaced now. If its cost were Rs 3200, keep it one more year.

5. Dynamic Programming (DP) - Introduction

DP is a method for solving **multistage (sequential) decision problems** by breaking the big problem into smaller sub-problems (stages), solving them one at a time and combining the results. It was developed by **Richard Bellman** (1950s).

Bellman's principle of optimality: An optimal policy has the property that, whatever the initial state and the initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.

DP terminology

- Stage: a point at which a decision is taken (period, activity, plant)
- State (s_n): information describing the situation at a stage (resources left, node reached)
- Decision variable (d_n) and return $r_n(s_n, d_n)$: choice made and the immediate gain/cost
- Transition: $s_{n-1} = T(s_n, d_n)$; Optimal value function: $f_n(s)$ = best total return from stage n to the end

Recursive (functional) equation: $f_n(s_n) = \text{opt over } d_n \{ r_n(s_n, d_n) + f_{n-1}(s_{n-1}) \}$

Backward recursion starts from the last stage and works back to the first (most common); forward recursion starts at stage 1.

Why the principle works (idea)

Suppose the best route from A to D passes through node B. Then the part of that route from B to D must itself be the best route from B to D - otherwise replacing it by a better B-to-D route would improve the whole route, a contradiction. So we may solve small problems from the end and build up the answer.

DP type	Meaning
Deterministic	state transitions and returns are known with certainty
Probabilistic (stochastic)	next state is random with known probabilities; expected values are optimised
Discrete / continuous	stages or states take discrete values / continuous values
Forward / backward	recursion from first stage / from last stage

Characteristics of a DP problem

- Problem divides into stages with a decision at each stage; each stage has a number of states
- The decision changes the state (transformation); the effect of the past is summarised in the present state (**Markov / memoryless** property)
- Principle of optimality holds; recursion gives the final answer

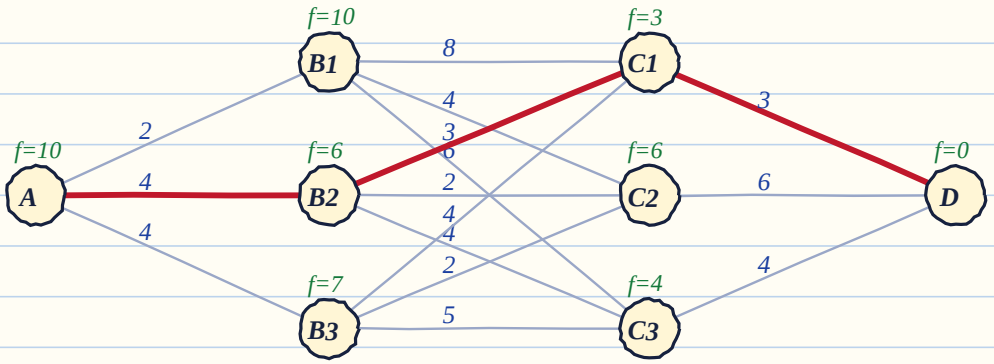
Applications

- Shortest/longest route, capital budgeting, resource allocation, inventory and production scheduling, equipment replacement, cargo loading (knapsack), LP problems

6. DP Application 1: Shortest Path Problem

Find the least-cost route from origin A (stage 0) to destination D through intermediate nodes.

Work **backwards**: f(node) = minimum of [arc cost + f(next node)].



Stage	Node	Choices (arc + f of next)	f(node)	Best next
3	C1 / C2 / C3	only D: 3, 6, 4	3 / 6 / 4	D
2	B1	8+3=11, 4+6=10, 6+4=10	10	C2 or C3
2	B2	3+3=6, 2+6=8, 4+4=8	6	C1
2	B3	4+3=7, 2+6=8, 5+4=9	7	C1
1	A	2+10=12, 4+6=10, 4+7=11	10	B2

Shortest path: A -> B2 -> C1 -> D with total length 4 + 3 + 3 = 10.

Forward check: f(A) = min(12, 10, 11) = 10. The red path in the figure is the optimal policy.

A second shortest-path example (table method)

Nodes 1-6 with arc lengths: 1-2: 4, 1-3: 2, 2-4: 5, 2-5: 3, 3-4: 1, 3-5: 6, 4-6: 2, 5-6: 2.

Backward:

Node	Choices	f(node)	Next
6	destination	0	-
5	2 + 0	2	6
4	2 + 0	2	6
3	1+2 = 3, 6+2 = 8	3	4
2	5+2 = 7, 3+2 = 5	5	5
1	4+5 = 9, 2+3 = 5	5	3

Shortest path 1 -> 3 -> 4 -> 6, length 2 + 1 + 2 = 5.

7. DP Application 2: Capital Budgeting Problem

A company has **Rs 5 crore** to allocate among three plants (stages). Returns $r_j(x)$ (Rs crore) from investing x crore (whole units) in plant j are given. Maximise total return.

Investment x	0	1	2	3	4	5
$r_1(x)$ Plant 1	0	3	7	9	12	13
$r_2(x)$ Plant 2	0	5	10	15	17	18
$r_3(x)$ Plant 3	0	5	6	8	9	10

Stage 1 and Stage 2

Stage 1: $f_1(x) = r_1(x)$ (all remaining money to Plant 1). Stage 2: $f_2(x)$ = max over k of [$r_2(k) + f_1(x - k)$].

State x	0	1	2	3	4	5
$f_1(x)$	0	3	7	9	12	13
$f_2(x)$	0	5	10	15	18	22
best k for plant 2	0	1	2	3	3	3

Example calc: $f_2(4)$ = max[0+12, 5+9, 10+7, 15+3, 17+0] = max[12, 14, 17, 18, 17] = 18 with k = 3. Stage 3: $f_3(5)$ = max over k of [$r_3(k) + f_2(5 - k)$]:

k (to plant 3)	0	1	2	3	4	5
$r_3(k) + f_2(5-k)$	0+22=22	5+18=23	6+15=21	8+10=18	9+5=14	10+0=10

Optimal allocation: Plant 3 gets 1 (return 5); remaining 4 -> Plant 2 gets 3 (return 15); remaining 1 -> Plant 1 gets 1 (return 3).

Maximum total return = 5 + 15 + 3 = Rs 23 crore. (backward tracing of the best k values)

8. DP Application 3: Cargo Loading (Knapsack) Problem

A vehicle of capacity 5 tonnes can carry three types of item (any number of each): weights 2, 3, 1 tonnes and values Rs 65, 80, 30. Maximise value. Stage = item type, state = capacity left.

Capacity x	0	1	2	3	4	5
$f_1(x)$: item 1 only	0	0	65	65	130	130
$f_2(x)$: items 1, 2	0	0	65	80	130	145
$f_3(x)$: items 1, 2, 3	0	30	65	95	130	160

Recursion: $f_j(x) = \max$ over $k \geq 0$ of $[\, k\, v_j + f_{j-1}(x - k\, w_j) \,]$. Tracing back from $f_3(5) = 160$: item 3 once (30, weight 1) and item 1 twice (130, weight 4). Maximum value = Rs 160.

Inventory/production example idea

Stage = month, state = stock at start of month, decision = production quantity, return = production + holding cost. Backward recursion $f_n(s) = \min [\, c(x) + h(s + x - d_n) + f_{n+1}(s + x - d_n) \,]$ gives the cheapest production plan.

9. DP Application 4: Solving an LP Problem by DP

Any LP with n variables can be treated as an n-stage problem: stage j chooses x_j; the **state** is the amount of each resource still unused.

Maximise $Z = 3x_1 + 5x_2$ s.t. $x_1 \leq 4$, $2x_2 \leq 12$, $3x_1 + 2x_2 \leq 18$, $x_1, x_2 \geq 0$.

- Stage 2 (x_2): state $s = 18 - 3x_1$ (capacity left in the 3rd constraint). $f_2(s) = \max 5x_2$ with $2x_2 \leq 12$ and $2x_2 \leq s \Rightarrow x_2 = \min(6, s/2)$, $f_2(s) = 5 \min(6, s/2)$
- Stage 1 (x_1): $f_1 = \max_{0 \leq x1 \leq 4} [\, 3x_1 + 5 \min(6, (18 - 3x_1)/2) \,]$
- For $x_1 \leq 2$: min = 6, value $= 3x_1 + 30$ (increasing). For $2 \leq x_1 \leq 4$: value $= 3x_1 + 5(9 - 1.5x_1) = 45 - 4.5x_1$ (decreasing)

Best at $x_1 = 2$: $x_2 = \min(6, 12/2) = 6$, $Z = 6 + 30 = 36$. Same as the simplex/graphical solution.

DP is useful for non-linear and integer problems too, but suffers from the **curse of dimensionality**: the number of states grows rapidly with the number of resources.

Point	Linear programming	Dynamic programming
Nature	single-shot optimisation	multi-stage, sequential
Method	simplex (standard algorithm)	no standard formula; recursion fitted to problem
Objective/constraints	must be linear	can be non-linear, integer, stochastic
Limit	very large m, n manageable	dimensionality problem

9. Summary Tables for Revision

Problem	Method / formula	Decision rule
Economic life (no interest)	$A(n) = [C - S_n + \sum R]/n$	min A(n); replace if $M(n+1) > A(n)$
Economic life (interest i)	$W(n)$ with $v = 1/(1+i)$	min $W(n)$
Sudden failure	$N_t = Np_t + \sum N_k p_{t-k}$	compare group C(t) vs individual cost
Make or buy	$Q_0 = F/(p - v)$	make if $Q > Q_0$
Shortest path (DP)	$f(i) = \min[d_{ij} + f(j)]$	backward from destination
Capital budgeting (DP)	$f_n(x) = \max[r_n(k) + f_{n-1}(x-k)]$	trace best k backwards

Typical exam steps

- Replacement: always tabulate year, running cost, cumulative cost, resale, total, average - the minimum average is the answer; verify with the marginal-cost rule
- Group replacement: compute failures per period using the renewal formula first; then group cost per period for t = 1, 2, ...; stop when cost starts to rise
- DP: define stages, states, decision, return and recursion **before** computing; show the table for each stage; trace back the optimal policy

Common mistakes

- Forgetting to subtract the resale value in A(n); using running cost instead of **cumulative** running cost
- In group replacement, counting the failures of the interval's last period as individual replacements (they are replaced as part of the group)
- In DP, working forward when the recursion was set up backward; not tracing back to get the actual decisions

10. Question Bank with Short Answers

2-mark questions

- Define economic life of an asset. -> The age at which the average annual cost of owning and running it is minimum.
- What is group replacement? -> Replacing all items together at fixed intervals, plus replacing individual failures in between.
- State Bellman's principle of optimality. -> Remaining decisions of an optimal policy must be optimal for the state reached after the first decision.
- What are the stages and states in DP? -> Stage = decision point; state = description of resources/position at that stage.
- Give two types of maintenance. -> Preventive (before failure) and breakdown/corrective (after failure).
- What is the curse of dimensionality? -> State space grows exponentially with the number of state variables, making DP tables huge.

5-mark questions

- Types of replacement problems with one example each; make-or-buy decision with break-even analysis.
- Explain the individual and group replacement policies and when group replacement is better.
- Formulate the shortest path problem as a DP and write the recursion.

10-mark questions

- Economic life from a table of running costs and resale values (with and without interest).
- Group replacement of items from a mortality table with cost comparison.
- Capital budgeting / cargo loading / shortest path by DP with full stage tables; solve a small LP by DP.

Study plan: first master the three table procedures (economic life table, renewal-equation table, stage table of DP). Practise each with two different data sets; marks are given for the layout of the table and the final decision statement.

11. Quick Revision & Important Questions

Replacement: $A(n) = [C - S_n + \sum R]/n$ minimum; replace when marginal cost > average cost. Group replacement: $C(t) = [N c_g + c_i \sum N_k]/t$; $N_t = N p_t + \sum N_k p_{t-k}$.
Make/buy: $Q_0 = F/(p - v)$. DP: $f_n(s) = \text{opt}\{ r + f_{n-1}(s') \}$; backward recursion; shortest path 10; capital budgeting 23.

Likely questions: (1) Types of maintenance; make-or-buy; (2) Types of replacement problems; (3) Find economic life of a machine from a cost table; (4) Individual vs group replacement for bulbs; (5) Bellman's principle with example; (6) Shortest path by DP; (7) Capital budgeting by DP; (8) Solve a small LP by DP.